

★ Branch and Bound

(1) FIFO

(2) LIFO

✓ (3) Least cost (LC)

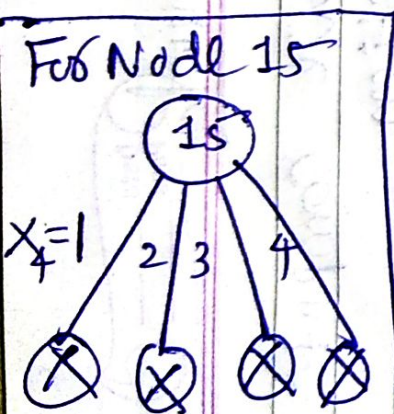
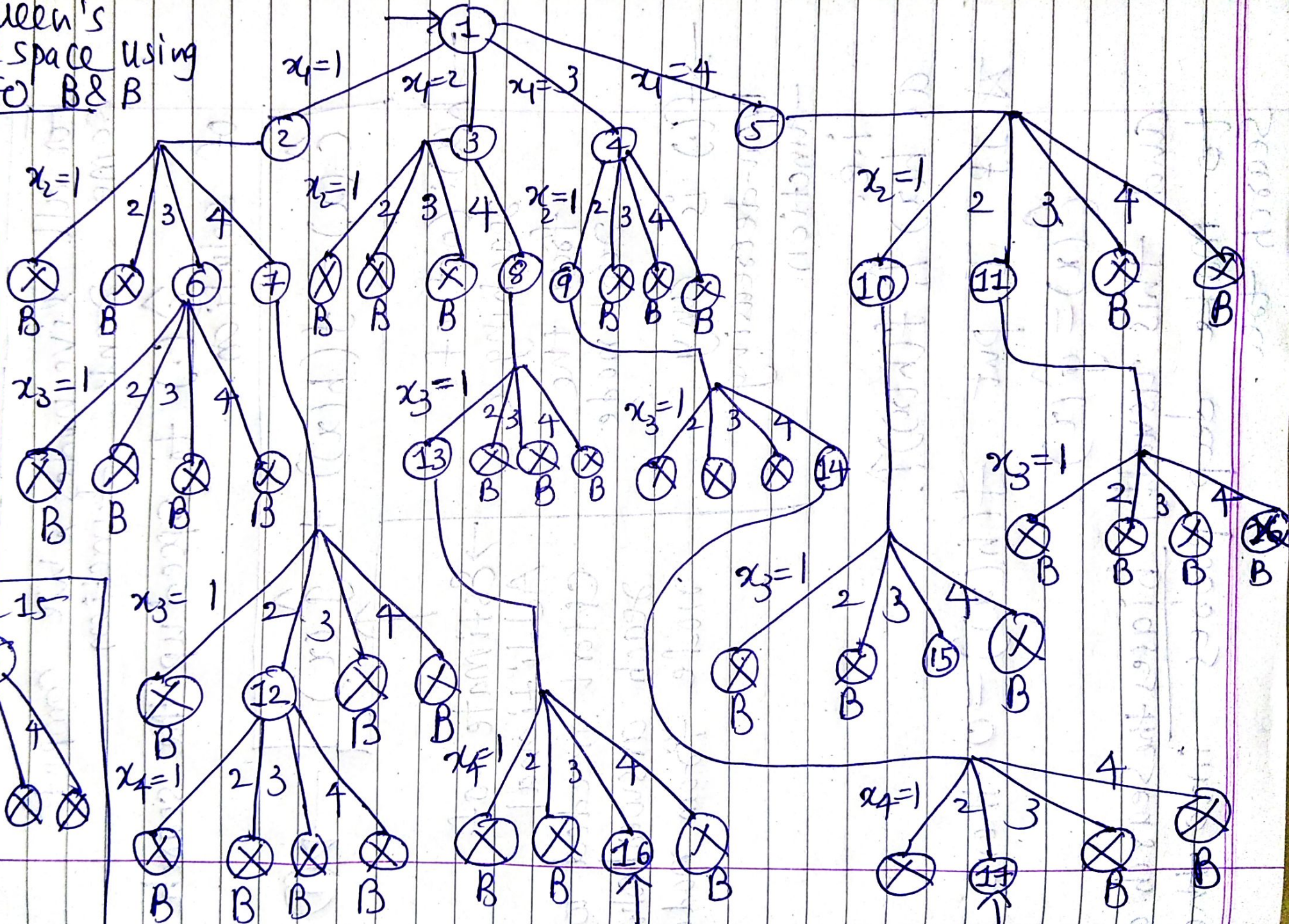


B & B

★ 4-Queen's State Space exploration
Using FIFO Branch & Bound.

4-Queen's State space using FIFO B&B

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So $\langle 2, 4, 1, 3 \rangle$

$\langle 3, 1, 4, 2 \rangle$

★ Least Cost Branch & Bound

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For all live nodes, we compute some rank function

$\hat{c}(\cdot)$ to speed up the search of solution.

$$\boxed{\hat{c}(x) = f(h(x)) + \hat{g}(x)} \rightarrow \textcircled{1}$$

→ $h(x)$: Cost of reaching x from the root node

→ $f(\cdot)$ is any non-decreasing function.

i.e.

as $h(x) \uparrow$, $f(h(x)) \uparrow$.

→ Estimate of Additional effort needed to reach answer/solution node from node x

★ If I put $f(h(x)) = 0$ then
 $\hat{c}(x) = \hat{g}(x)$

and this usually biases the search algo. to make deep probes into the search tree.

↓
Why?

because $\hat{g}(y) \leq \hat{g}(x)$

where y is a child of x

→ A Search strategy that uses a Cost function

$$\hat{C}(x) = f(h(x)) + \hat{g}(x)$$

to select the next E-node would always choose for its next E-node a live node with least $\hat{C}(\cdot)$.



This is LC-Search.

★ B&B is used to solve Minimization types of problems.



i.e. Optimization problems

Ex: 0/1 Knapsack

Job Assignment

Traveling Salesman Problem

8-puzzle